

Certified Integral Properties of the Dipper Block Cipher: Monomial-Trail Analysis and the Role of the Half-State Modular Addition

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Abstract

Dipper is a 64-bit lightweight block cipher whose round combines the GIFT S-box and bit permutation with two 16-bit modular additions on half of the state. Its integral resistance was so far supported only by experiments with a few random keys, which bounded word-saturation distinguishers at five rounds. We replace this evidence by key-independent certificates. We model monomial trails through reduced-round Dipper as SAT instances whose local transitions are exact: the coefficient table of the S-box and a rule for the map $(x, y) \mapsto (x \boxplus y, y)$ derived from the Braeken–Semaev characterisation of modular addition. The certificates at the round boundaries are backed by DRAT proofs checked with `drat-trim`, and every trail behind a negative boundary result is validated independently of the SAT encoding. The five-round properties become statements for all round keys, and six-round properties exist. Within this existence-based model, no bit-aligned cube certifies any single output bit at seven rounds; a monotonicity lemma makes this statement cover cubes of every dimension. A controlled ablation isolates the effect of the two modular additions: replacing them by XOR, or removing them, extends the longest certified property from six rounds to nine and ten rounds. At the six-round boundary every certified bit comes from the two words that bypass the addition, and we relate this to the exact degree of the addition output bits and to degree upper bounds for every output bit and round. On all instances examined, the model with exact local transitions certifies the same bits as the conventional bit-based division property.

Keywords: integral cryptanalysis; division property; monomial prediction; modular addition; SAT; DRAT; lightweight block cipher; Dipper

1 Introduction

Integral (square, saturation) cryptanalysis [24] looks for a set of chosen plaintexts, typically an affine subspace called a *cube*, over which some output bit sums to zero independently of the key. The division property [32] and its bit-based versions [34, 38] turned the search for such properties into a propagation problem for MILP or SAT solvers. Monomial prediction [17], which Hu et al. proved equivalent to the three-subset division property without unknown subset [14], gives the exact algebraic view: a cube sum vanishes if and only if certain monomial coefficients are zero.

Modular addition is well understood in isolation. Its differential and linear behaviour is described exactly by Lipmaa and Moriai [27] and by Wallén [35], and its algebraic normal form by Braeken and Semaev [8]. Integral analysis of ciphers that mix modular addition with S-boxes is less developed. Division-property models for ARX designs [30, 29] target pure ARX ciphers

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such as SPECK [2], in which the addition is the only nonlinear operation. In a hybrid round the addition interacts with a small S-box and a bit permutation. It is then not obvious how much the addition contributes to integral resistance, or which part of the state keeps an integral property longest.

Dipper [21] is a 64-bit hybrid SPN-ARX cipher with 96- and 128-bit keys and 28 rounds, proposed in our previous work. Each round applies a full-state key addition, sixteen GIFT S-boxes [1], four word rotations, two 16-bit modular additions over half of the state and the GIFT-64 bit permutation. The design paper evaluates differential, linear and impossible-differential resistance with MILP and CP-SAT models. Its integral evaluation, however, is *experimental*: one 16-bit word is saturated, the sums are tested for a few random keys, and distinguishers are reported for up to five rounds. Such an evaluation has two weaknesses. A sum that is zero for a few random keys need not be zero for all keys, and a search limited to word-saturated cubes says nothing about other cubes. Section 6 shows that the first weakness is not hypothetical for Dipper: with 80 trials, two bits look balanced that are not.

Research questions. We ask three questions. (Q1) Which integral properties of reduced-round Dipper hold for *all* round keys, and how far do they extend over *all* bit-aligned cubes? (Q2) What exactly do the two half-state additions contribute, and why do some parts of the state stay balanced longer than others? (Q3) Does an exact treatment of the modular addition, made possible by the Hu-Yap characterisation [20], certify more than the conventional division property?

Contributions.

1. **Key-independent certificates and a complete search within the model (Q1).** We certify the published five-round properties for all round keys, find six-round properties (nine balanced bits with a 2^{63} cube; six bits with the smallest certified cube found, of size 2^{60}), and show that no bit-aligned cube certifies a single output bit at seven rounds in the model. The monotonicity lemma (Lemma 3) reduces this statement to the 64 cubes of dimension 63. Every boundary result has a checkable artifact: 138 DRAT proofs (distinct instances) verified by `drat-trim`, and 12 288 monomial trails validated by a checker that does not use the SAT encoding.
2. **The role of the modular addition (Q2).** In a controlled ablation, replacing the two additions by XOR or removing them extends the longest certified property from six to nine and ten rounds. At the six-round boundary every certified bit comes from the retained operands, the two words that feed the additions but pass through them unchanged. We relate this to Lemma 2, which states that output bit i of an addition has degree exactly $i + 1$, and to degree upper bounds for every output bit and round.
3. **Model comparison and tightness (Q3).** The monomial-trail existence model with exact local transitions (MP-EL), a gate-level monomial model and the conventional bit-based division property certify the same bits on every instance we examined. We explain part of this structurally, and we measure the distance between certificates and experiment.

The technical tools behind these results are the exact local rule for Dipper’s ARX map (Lemma 1) and the free-final-round observation (Lemma 4). Sections 9 and 10 contain secondary material: a partial-key filtering procedure and a specification check. Code, data, proofs and scripts are available at <https://github.com/AlexGuseinov/dipper-integral>.

Scope of the claims. A certificate proves that a cube sum is zero for all round keys. The absence of a certificate proves nothing about the cipher: it only means that the model contains a trail. Our seven-round statement is therefore a statement about the model, not a proof that

Dipper has no seven-round integral distinguisher. For cipher outputs we report only degree upper bounds and never claim that a degree is attained; Lemma 2, about an isolated addition, is the only exact degree statement.

Organisation. Section 2 reviews related work and Section 3 introduces the notation and background. Section 4 gives the local rules and the structural lemmas, and Section 5 validates them. Section 6 reports the certified properties of Dipper, and Section 7 studies the role of the addition. Section 8 compares models and experiment. Sections 9 and 10 cover key recovery and the specification check. Section 11 discusses implications and limitations.

2 Related work

Integral and higher-order differential cryptanalysis. Integral attacks go back to the Square attack [10]. They were formalised as integral cryptanalysis by Knudsen and Wagner [24] and as the saturation attack by Lucks [28], and extended to bit-oriented ciphers by Z’aba et al. [39]. The underlying algebra is that of higher-order derivatives [26, 23]: a cube sum is a derivative whose order is the cube dimension, and it vanishes whenever the algebraic degree is smaller than that order. Canteaut and Videau [9] and Boura, Canteaut and De Cannière [7] bounded the degree of iterated constructions. The cube attack of Dinur and Shamir [11] applies the same algebra to key recovery.

Division property and monomial prediction. Todo introduced the division property [32] and used it to break full MISTY1 [31]. The bit-based division property [34] and its MILP modelling [38] made automated search practical, and Eskandari et al. [12] provide a SAT-based tool and a table of results that we use as a benchmark. In stream ciphers the division property underlies modern cube attacks [33, 36]. The three-subset division property [18] and its variant without unknown subset [14] remove the imprecision of the two-subset version; Hu et al. [17] showed that the latter is equivalent to monomial prediction. Complex linear layers need dedicated models [19, 41].

Modular addition. Sun, Wang and Wang [30] gave the first MILP model of the bit-based division property for ARX ciphers, based on a gate-level decomposition of the carry chain, and Sun et al. [29] moved it to SAT. Hu and Yap [20] observed that the characterisation of Braeken and Semaev [8] yields a perfect local monomial-prediction model for modular addition. CLAASP-MP [3] integrates such models into a general MILP framework. We use this characterisation for the specific map $(x, y) \mapsto (x \boxplus y, y)$ of Dipper and derive from it a degree statement.

Security guarantees and degree lower bounds. Finding an integral distinguisher answers an upper-bound question. Proving that *no* integral distinguisher exists requires exact information, such as lower bounds on the degree. Hebborn et al. developed degree lower bounds [15] and strong security guarantees against integral distinguishers [16] for ciphers in which a key is added to the full state; Dipper, which adds a full-state key before its first S-box layer, appears to fit this setting. Zeng and Tian [40] extended the guarantees to ciphers without such a whitening key, such as SIMON and Simeck.

Integral key recovery. Hadipour and Eichlseder [13], for example, study key recovery built on monomial prediction.

Positioning. We do not propose a new general framework. Our contribution is the analysis of one hybrid cipher, which consists of exact rules for its particular ARX shape, key-independent

certificates with checkable artifacts, a search over all bit-aligned cubes within the model, a model comparison, an ablation, and a degree-based explanation.

3 Preliminaries

This section fixes notation and recalls the algebraic facts on which the analysis rests. The material is standard, but we state it in the form in which we use it, because the difference between what a model *proves* and what it merely *fails to exclude* runs through the whole paper.

3.1 Notation

Vectors in $\mathbb{F}_2^n$ are written $u = (u_0, \dots, u_{n-1})$ with u_0 the least significant bit. We identify u with the integer $\text{val}(u) = \sum_i u_i 2^i$ whenever we do arithmetic with it. The Hamming weight is $\text{wt}(u) = \#\{i : u_i = 1\}$, and e_i is the i -th unit vector. We write $u \preceq v$ if $u_i \leq v_i$ for all i , so that u is a sub-mask of v ; $u \vee v$ and $u \wedge v$ are the bitwise OR and AND. For a set $I \subseteq \{0, \dots, n-1\}$, $\mathbf{1}_I$ is its indicator vector. The symbol $\oplus$ denotes XOR, and $\boxplus$ addition modulo 2^n , where n is clear from the context. $x \lll s$ is a left rotation of an n -bit word.

3.2 Boolean functions, ANF and degree

Every Boolean function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ has a unique algebraic normal form (ANF)

$$f(x) = \bigoplus_{u \in \mathbb{F}_2^n} a_u x^u, \quad x^u = \prod_{i: u_i=1} x_i, \quad a_u \in \mathbb{F}_2.$$

The coefficients follow from the truth table by the Möbius transform $a_u = \bigoplus_{v \preceq u} f(v)$, which costs $n2^n$ operations. The *algebraic degree* $\deg f$ is the largest $\text{wt}(u)$ with $a_u = 1$. For a vectorial function $F = (f_0, \dots, f_{m-1})$ and $v \in \mathbb{F}_2^m$ we write $F^v = \prod_{j: v_j=1} f_j$, which is again a Boolean function. Two facts are used repeatedly: $x_i^2 = x_i$, so products of monomials are monomials of the union of their supports; and a nonconstant component of a permutation of $\mathbb{F}_2^n$ has degree at most $n-1$, since $\bigoplus_x f(x)$ equals the coefficient $a_{\mathbf{1}}$ and is zero for a balanced function.

3.3 The Dipper cipher

Figure 1 shows one round. The state is $S = A\|B\|C\|D$ with 16-bit words, $A = S[63:48]$, $D = S[15:0]$ and bit 0 the least significant. For $r = 1, \dots, 28$,

$$S^{(r)} = T(S^{(r-1)} \oplus RK_r), \quad T = P \circ M \circ SC,$$

with $S^{(0)}$ the plaintext and $S^{(28)}$ the ciphertext. SC applies the GIFT S-box to the sixteen nibbles. M first rotates A, B, C, D left by 1, 4, 7, 11 and then sets

$$A \leftarrow A \boxplus B, \quad C \leftarrow C \boxplus D,$$

so B and D are used as addends and pass through the additions unchanged (after their rotation). We call A, C the *added* words and B, D the *retained* words of the ARX layer. P is the GIFT-64 bit permutation $P(i) = 4\lfloor i/16 \rfloor + 16((3\lfloor (i \bmod 16)/4 \rfloor + (i \bmod 4)) \bmod 4) + (i \bmod 4)$, which moves bit i to position $P(i)$. Three structural features matter for integral analysis. The key is added to the *full* state at the start of every round. T is a public permutation. There is no key addition after the last round.

For the ablation we define two analysis variants, which are not ciphers anyone proposes. $\text{Dipper}^\oplus$ replaces both additions by XOR, and Dipper° removes them while keeping the rotations. Throughout, the round keys $RK_1, \dots, RK_r$ are treated as *independent* variables; a statement that holds for all such sequences holds in particular for the sequences produced by either key schedule.

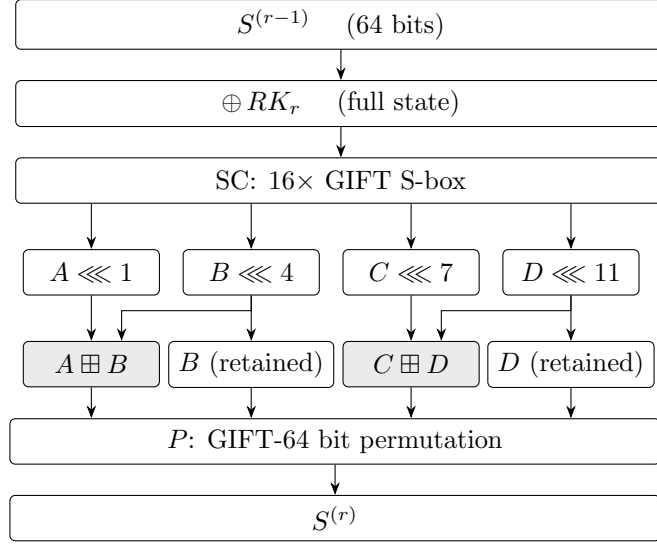


Figure 1: One Dipper round. The shaded boxes are the *added* words; B and D are the *retained* words, used as addends and passed on unchanged. The ARX layer M consists of the rotations and the two additions. There is no key addition after the last round.

3.4 Cubes and integral properties

Definition 1 (Cube, balanced bit). Let $I \subseteq \{0, \dots, 63\}$ and $c \in \mathbb{F}_2^{64}$ with $c_i = 0$ for $i \in I$. The *cube* $\mathcal{C}_I(c)$ is the set of the $2^{|I|}$ plaintexts that take all values on the positions in I (the active bits) and agree with c elsewhere (the constant bits). For an r -round state bit $S_j^{(r)}$, the *cube sum* is $\bigoplus_{x \in \mathcal{C}_I(c)} S_j^{(r)}(x)$. The bit is *balanced over I after r rounds* if the cube sum is zero for every constant c and every sequence of round keys.

All cubes in this paper are *bit-aligned*: they are spanned by unit vectors. Affine subspaces in other bases, non-affine input sets and output functions other than a single bit are outside the scope of our search.

Viewing $S_j^{(r)}$ as a polynomial in the plaintext bits x and the key bits k , and substituting the constants c_i for the plaintext bits outside I after summation,

$$\bigoplus_{x_I \in \mathbb{F}_2^{|I|}} S_j^{(r)}(x, k) = \bigoplus_{u \succeq \mathbf{1}_I, v} a_{u,v} x^{u \oplus \mathbf{1}_I} k^v, \quad (1)$$

because summing a monomial x^u over the cube leaves $x^{u \oplus \mathbf{1}_I}$ if $u \succeq \mathbf{1}_I$ and zero otherwise. Hence the bit is balanced over I *if and only if* every coefficient $a_{u,v}$ with $u \succeq \mathbf{1}_I$ vanishes. A cube sum is a derivative of order $|I|$ [26]. In particular, if $\deg_x S_j^{(r)} < |I|$ the bit is balanced over every cube of dimension $|I|$, which links integral properties to degree bounds (Section 7.2). The 64-dimensional cube, i.e. the full codebook, is balanced for every bit of every permutation. It is trivial and excluded.

3.5 Monomial trails

Let $F = F_r \circ \dots \circ F_1$ be a composition of vectorial functions. For a single function G and masks u, v , write $G[u \rightarrow v] = 1$ if x^u occurs in the ANF of G^v and 0 otherwise. A *monomial trail* from u_0 to u_r is a sequence $u_0 \rightarrow u_1 \rightarrow \dots \rightarrow u_r$ with $F_t[u_{t-1} \rightarrow u_t] = 1$ for all t .

Proposition 1 (Hu et al. [17]). *The coefficient of x^{u_0} in F^{u_r} equals the number of monomial trails from u_0 to u_r modulo 2.*

The proposition gives two sound tests of opposite strength. If *no* trail exists, the coefficient is zero; this is the test we use, and it only needs a satisfiability check. If trails exist, the coefficient is their parity. Deciding it requires counting, and trails can cancel in pairs. The following toy example shows the gap between the two tests.

Example 1 (Cancellation). Let $F_1(x_1, x_2) = (x_1x_2, x_1x_2)$ and $F_2(y_1, y_2) = y_1 \oplus y_2$. Then $F = F_2 \circ F_1 = 0$, so the sum over the cube $\{x_1, x_2\}$ is zero. From $x^{11} = x_1x_2$ to the output there are two trails, $11 \rightarrow 10 \rightarrow 1$ and $11 \rightarrow 01 \rightarrow 1$, because x_1x_2 occurs in both y_1 and y_2 and both occur in F_2 . An existence test finds a trail and cannot certify the property. The parity test counts two trails and certifies it.

For keyed functions we treat each key bit as a variable. The key-addition layer $s' = s \oplus k$ satisfies $s'^w = \bigoplus_{u \preceq w} s^u k^{w \oplus u}$. A trail step $u \rightarrow w$ is therefore possible exactly when $u \preceq w$, and different steps attach different key monomials $k^{w \oplus u}$. Trails with different key monomials never cancel each other; only trails with the same key monomial can.

3.6 The bit-based division property

The conventional (two-subset) bit-based division property [34] describes a multiset $\mathbb{X} \subseteq \mathbb{F}_2^n$ by a set $\mathbb{K}$ of vectors: $\mathbb{X}$ has property $\mathcal{D}_{\mathbb{K}}^n$ if $\bigoplus_{x \in \mathbb{X}} x^u = 0$ for every u such that $u \not\preceq k$ for all $k \in \mathbb{K}$. A cube with active set I has $\mathbb{K} = \{\mathbf{1}_I\}$. Propagation rules map $\mathbb{K}$ through each operation, and bit j is balanced if e_j is not reachable. Key addition leaves the property unchanged. For an S-box, $k \rightarrow v$ is a *division trail* if some $u \succeq k$ has $S[u \rightarrow v] = 1$. The rules for COPY, AND and XOR gates are those of [34, 38]. BDP is sound in the same direction as the existence test above. Like that test, it cannot see cancellations, and it can be less precise still. Precision is improved by the three-subset division property [18] and made exact by its variant without unknown subset [14], which is equivalent to monomial prediction [17].

3.7 SAT encodings and checkable answers

Both tests become satisfiability problems once every mask bit is a Boolean variable and every local relation is written as clauses. Each clause excludes one forbidden assignment of the variables of a small component. The solver then answers one of two ways. If the formula is *unsatisfiable* (UNSAT), no trail exists and the bit is certified; a DRAT proof [37] records the solver’s reasoning, and an independent checker such as `drat-trim` can replay it. If the formula is *satisfiable* (SAT), the solver returns an assignment, from which we read off a candidate trail and verify it layer by layer against the local rules, without using the clauses. We use CaDiCaL [5, 4] through PySAT [22] for the search and an external CaDiCaL binary for proof generation.

4 Models

4.1 Local propagation rules

Key addition. A step $u \rightarrow w$ is allowed iff $u \preceq w$. Under BDP the key addition is transparent.

S-box. For the GIFT S-box we compute $S[u \rightarrow v]$ for all $u, v \in \mathbb{F}_2^4$ by the Möbius transform. Of the 256 pairs, 68 are nonzero. The BDP table has 170 transitions.

Modular addition with retained operand. Braeken and Semaev [8], and Hu and Yap [20] in the cryptanalytic setting, show that for $z = x \boxplus y$ on n bits the monomial $x^a y^b$ occurs in z^w if and only if $\text{val}(a) + \text{val}(b) = \text{val}(w)$ over the integers. Dipper’s ARX map also outputs the addend. The following lemma gives the exact rule for that shape.

Lemma 1. *Let $F(x, y) = (x \boxplus y, y)$ on n -bit words. Then $[x^a y^b] (x \boxplus y)^w y^c = 1$ if and only if $\text{val}(w) \geq \text{val}(a)$ and $b = b' \vee c$, where b' is the word with $\text{val}(b') = \text{val}(w) - \text{val}(a)$.*

Proof. By the Braeken–Semaev characterisation, $(x \boxplus y)^w = \sum_{\text{val}(a') + \text{val}(b') = \text{val}(w)} x^{a'} y^{b'}$. Multiplying by y^c and using $y_i^2 = y_i$ gives $\sum x^{a'} y^{b' \vee c}$. The coefficient of $x^a y^b$ is the parity of $\#\{b' : \text{val}(a) + \text{val}(b') = \text{val}(w), b' \vee c = b\}$. The equation determines b' uniquely, and b' exists iff $\text{val}(w) \geq \text{val}(a)$. $\square$

Example 2. For $n = 2$, the high output bit of $z = x \boxplus y$ is $z_1 = x_1 \oplus y_1 \oplus x_0 y_0$. Its monomials correspond to the pairs $(\text{val}(a), \text{val}(b)) \in \{(2, 0), (0, 2), (1, 1)\}$, which are exactly the solutions of $\text{val}(a) + \text{val}(b) = 2 = \text{val}(e_1)$. The monomial $x_0 y_0$ comes from the carry, and its degree $2 = i + 1$ anticipates Lemma 2 below. For the retained version, $z_1 y_0 = x_1 y_0 \oplus y_1 y_0 \oplus x_0 y_0$. With $w = e_1$ and $c = e_0$, Lemma 1 predicts the monomials $x^a y^{b' \vee e_0}$ for the same three pairs, i.e. $x_1 y_0$, $y_1 y_0$ and $x_0 y_0$, in agreement with the direct computation.

The same characterisation gives the exact degree of each output bit of the addition. Lemma 2 is an elementary consequence of [8]. We state it because it drives the mechanism of Section 7.2.

Lemma 2 (Degree of an addition bit). *Let $z = x \boxplus y$ on n -bit words. For $0 \leq i < n$, the output bit z_i has algebraic degree exactly $i + 1$ as a Boolean function of the $2n$ input bits.*

Proof. Take $w = e_i$, i.e. $\text{val}(w) = 2^i$. The monomial $x^a y^b$ occurs in z_i iff $\text{val}(a) + \text{val}(b) = 2^i$, and it has degree $\text{wt}(a) + \text{wt}(b)$. Every carry in the binary addition of a and b turns two ones into a single one in the next position, so $\text{wt}(a) + \text{wt}(b) = \text{wt}(a + b) + c(a, b)$, where $c(a, b)$ is the number of carries (equivalently, by Kummer’s theorem [25], $c(a, b)$ is the 2-adic valuation of $\binom{a+b}{a}$). Here $\text{wt}(a + b) = 1$, and carries can occur only at positions $0, \dots, i - 1$, so the degree is at most $i + 1$. The pair $a = 1, b = 2^i - 1$ produces i carries and attains $i + 1$. $\square$

Corollary 1. *In one Dipper round, let d be an upper bound on the degree of the round input. Every bit of the retained words after the ARX layer is an S-box output bit and has degree at most $3d$. Bit i of an added word is a polynomial of degree $i + 1$ in S-box output bits, so its degree is at most $\min\{3d(i + 1), 63\}$. Only bit 0 of an added word is linear in the S-box outputs.*

Corollary 1 gives upper bounds; it does not state which degrees are attained.

Encoding. In the SAT model the equation $\text{val}(a) + \text{val}(b') = \text{val}(w)$ is encoded bitwise with auxiliary carries, $a_i + b'_i + q_i = w_i + 2q_{i+1}$, $q_0 = q_n = 0$. These are carries of the exponent integers, not the cipher’s data carries. The pair (a, w) determines b' and q , so each trail has exactly one extension to the auxiliary variables.

The three models. MP-EL (monomial-trail existence model with exact local transitions) uses the S-box table and Lemma 1. Its local relations are exact, but its global test is trail *existence*, not trail parity, so it cannot see cancellations of the kind in Example 1. For comparison, MP-circuit models the addition as a ripple-carry circuit, $s_i = x_i \oplus y_i \oplus c_i$ and $c_{i+1} = x_i y_i \oplus (x_i \oplus y_i) c_i$, with explicit COPY gates, including the copy of y that forms the retained output. It uses the monomial gate rules (COPY: $u = \bigvee v_j$; AND: $u_1 = u_2 = v$; XOR: $v = u_1 + u_2$). BDP uses the same circuit with the division-property rules (COPY: $u = \sum v_j$; AND: $v = u_1 \vee u_2$) and the BDP S-box table.

4.2 Round model, soundness and structural lemmas

One round is modelled as the key addition, sixteen S-box relations, the word rotations (wiring), the two addition relations and the bit permutation (wiring). The input mask is fixed to 1_I and the output mask after r rounds to e_j . Each output bit is an incremental query under assumptions.

Proposition 2 (Soundness). *If MP-EL (or BDP) is unsatisfiable for (I, r, j) , then bit j of $S^{(r)}$ is balanced over I : the cube sum is zero for every constant and every sequence of round keys, and in particular under both Dipper key schedules.*

Proof. Every local relation contains all transitions with nonzero coefficient. By Proposition 1, every nonzero coefficient $a_{u,v}$ with $u \succeq \mathbf{1}_I$ in (1) therefore has at least one trail starting from u . Such a trail first passes the key addition to some $w \succeq u \succeq \mathbf{1}_I$. Since $\mathbf{1}_I \preceq w$ is also a valid key-addition step, the model, whose input mask is exactly $\mathbf{1}_I$, contains a trail too. For BDP the statement is the standard soundness of division-property propagation [34, 38]. The round keys are independent variables and the constants merge with RK_1 , which gives the claimed generality. $\square$

Lemma 3 (Monotonicity). *In MP-EL, if $I \subseteq I'$ and (I, r, j) has no trail, then (I', r, j) has no trail. Consequently, if no bit-aligned cube of dimension 63 is certified for output bit j after r rounds, no bit-aligned cube of any dimension ≤ 63 is.*

Proof. The first operation is the key addition, so the masks reachable from $\mathbf{1}_I$ after it are $\{w \succeq \mathbf{1}_I\}$. This set contains $\{w \succeq \mathbf{1}_{I'}\}$, and all later constraints are identical. Every cube of dimension at most 63 lies inside some cube of dimension 63. $\square$

Lemma 4 (Free final round). *If bit j of $S^{(r)}$ is balanced over I and $|I| \geq 1$, then bit j of $T^{-1}(S^{(r+1)})$ is balanced over I .*

Proof. $T^{-1}(S^{(r+1)}) = S^{(r)} \oplus RK_{r+1}$, and the constant RK_{r+1} cancels over the even number $2^{|I|}$ of terms. $\square$

Lemma 4 uses a different output test: the ciphertext is first passed through the public map T^{-1} . It gives no information about RK_{r+1} . We report round counts for $S^{(r)}$ and state the $(r+1)$ -round extension separately.

Remark 1 (Degree bounds versus cubes). A bound $\deg_x S_j^{(r)} \leq 62$ certifies bit j over every 63-dimensional cube. The converse direction is weaker. Even a proof that $\deg_x S_j^{(r)} = 63$ would exclude a property for only those cubes I whose monomial $x^{\mathbf{1}_I}$ actually occurs. Excluding all 63-dimensional cubes would require the presence of the monomial $x^{\mathbf{1}_I}$ for each of the 64 choices of I separately.

5 Validation

Reference implementation. Our Python implementation reproduces both published Dipper-64/128 test vectors and is cross-checked against a vectorised implementation. Section 10 discusses the key schedule.

Local models. The S-box table is computed from the ANF. We verified exhaustively the Hu–Yap rule for $n \leq 5$, Lemma 1 for $n \leq 4$ (all 2^{16} exponent pairs), Lemma 2 for $n \leq 8$, and the carry encoding for $n \leq 6$.

Benchmark. With the same SAT machinery and BDP rules we reproduce the results of Eskandari et al. [12] for GIFT-64 [1] and PRESENT [6]:

- GIFT-64: a 9-round property with 63 active bits and no 10-round property. When the constant bit is the most significant bit of a nibble, as in the published result, we find 30 balanced bits; the best position gives 32.
- PRESENT: a 9-round property with 60 active bits and one balanced bit, and no 10-round property.

Soundness against experiment. We evaluated cube sums for random constants and independent random round keys in three settings:

- the word cubes of Table 1, with 1016 trials;
- the 28 cubes of the tightness study in Section 8, with 200 trials per cube and round;
- every cube of dimension at most 16 in Table 3, including those of the variants, with 400 trials.

No certified bit ever had a nonzero sum.

Checkable artifacts. For the boundary results listed below we produced artifacts that can be verified without trusting the SAT solver. The correctness of the CNF generator itself is not covered by these artifacts; it rests on the exhaustive checks of the local rules above and on the agreement with experiment. For the following UNSAT answers (certificates), the CNF was resolved by an external CaDiCaL 3.0.1 [4] binary with a DRAT proof, and the proof was checked by `drat-trim` [37]:

- every certified (cube, bit) pair of Dipper at six rounds;
- every such pair of $\text{Dipper}^\oplus$ at nine rounds and of $\text{Dipper}^\emptyset$ at ten rounds;
- the smallest cubes of Table 3 for Dipper;
- the five-round word-cube properties.

Each SAT answer (no certificate) was converted into a trail and validated by a checker that re-derives every layer from the cipher specification and the local rules, without the CNF. We did this for every one of the 3×4096 (cube, bit) pairs at Dipper seven rounds, $\text{Dipper}^\oplus$ ten rounds and $\text{Dipper}^\emptyset$ eleven rounds. All 141 proof runs, covering 138 distinct instances (three five-round instances occur both as word cubes and as frontier cubes), were verified, and all 12 288 trails passed the check. The other UNSAT answers, e.g. the certificates at $r \leq 5$ in Table 2, the comparison instances and the degree bounds, rely on the solver answer alone. The CNF hashes, the tool versions and the verification log are part of the repository.

6 Certified properties of Dipper

6.1 Reproducing the published experiment

Table 1 repeats the published experiment: one 16-bit word is saturated, the other 48 bits are constant, and 1016 trials are run (1000 with independent random round keys, 16 with the real 128-bit key schedule). A bit counts as empirically balanced only if its sum is zero in every trial. As published, five rounds are the maximum, and the round count refers to the state $S^{(r)}$ itself. For every word and every round $r = 2, \dots, 5$, MP-EL certifies exactly the empirically balanced bits, so these properties now hold for all round keys. The number of trials matters. An earlier run with 80 trials reported two more bits for word C , one at $r = 4$ and one at $r = 5$, whose sums turned out to be nonzero with probability of roughly 0.5–1%. Certificates remove exactly this kind of error. In every trial we also confirmed $\bigoplus T^{-1}(S^{(r+1)}) = \bigoplus S^{(r)}$ (Lemma 4).

6.2 Certified properties over all bit-aligned cubes

By Lemma 3, the 64 cubes of dimension 63 (one constant bit p) decide which rounds admit a certificate for some bit-aligned cube. Table 2 summarises them. At six rounds, certificates exist for 14 of the 64 positions of the constant bit, and the best positions $p \in \{4, 6, 7\}$ give nine

Table 1: Balanced bits of $S^{(r)}$ for 16-bit word-saturated cubes. For Dipper, the empirically zero bits (1016 trials) coincide with the certified bits in every entry for $r \geq 2$. The last two columns give, for the analysis variants, the last round with empirically zero bits (400 trials) and their number (416 trials: 400 with random round keys, 16 with the real key schedule); these two columns are empirical only. Bold: last round with balanced bits.

Saturated word	$r=1$	2	3	4	5	6	Dipper [⊕]	Dipper [∅]
<i>A</i>	64	64	64	38	3	0	6 (26 bits)	7 (14 bits)
<i>B</i>	64	56	24	0	0	0	5 (7 bits)	7 (9 bits)
<i>C</i>	64	64	64	38	3	0	6 (25 bits)	7 (4 bits)
<i>D</i>	64	60	25	0	0	0	5 (8 bits)	7 (6 bits)

Table 2: Cubes of dimension 63 (constant bit p ; 64 cubes per row). MP-EL, MP-circuit and BDP give identical results (MP-circuit was run for $r \leq 7$).

Rounds r	cubes with a certificate	max. balanced bits	solver time (MP-EL, 64 cubes)
4	64	64	3.9 s
5	64	42	6.1 s
6	14	9	7.9 s
7	0	0	8.6 s
8	0	0	11.1 s
9	0	0	12.2 s

balanced bits, $\{1, 18, 19, 24, 41, 48, 51, 56, 58\}$. At seven, eight and nine rounds no position gives a certificate. Within the model, therefore, no bit-aligned cube of any dimension certifies a single output bit at seven rounds or more. Every one of the 4096 seven-round queries returned a trail that passed the independent check. With Lemma 4, the six-round properties give seven-round distinguishers on $T^{-1}(C)$.

Data complexity. Starting from certified cubes, we removed active bits one at a time as long as some bit stayed certified, restarting from several cubes and random orders. Table 3 lists the smallest certified cubes *found* in this way. They are upper bounds on the data required, not proven minima. At five rounds the smallest cube found has size 2^{16} , the size of the published word-saturation structure (word C). At six rounds it has size 2^{60} : it keeps the nibble $\{4, 5, 6, 7\}$ constant and certifies six bits.

6.3 Where the balanced bits come from

We map the nine balanced bits at the six-round boundary back through the final bit permutation. Every one of them is an output of a retained word in the last round: bits $D_3, D_4, D_5, D_6, D_{11}$ and $B_4, B_{12}, B_{13}, B_{14}$. The same holds for the five-round word-cube properties. Bits $\{51, 56, 58\}$ of the word- C cube map to D_3, B_4, B_{14} , and bits $\{1, 19, 24\}$ of the word- A cube map to D_5, D_{11}, B_{12} . No output bit of the last-round additions is certified at the boundary. This is an observation about the certified bits, not a claim about every output bit of the additions: bit 0 of $A \boxplus B$, for example, is linear. Section 7.2 relates it to degree.

Table 3: Smallest certified cubes found (dimension d , i.e. 2^d chosen plaintexts; greedy search, not proven minimal). “–”: no certificate for any bit-aligned cube (Lemma 3); blank: not computed.

Rounds r	3	4	5	6	7	8	9	10
Dipper	2	4	16	60	–	–	–	
Dipper [⊕]			7	22	44	59	63	–
Dipper [∅]			3	8	15	48	59	63

Table 4: Longest round count with a certificate for some bit-aligned cube (all cubes, by Lemma 3) and the maximum number of balanced bits at that round.

Variant	longest certified r	max. balanced bits	with free round (Lemma 4)
Dipper (⊞)	6	9	7
Dipper [⊕]	9	5	10
Dipper [∅]	10	2	11

7 The role of the modular addition

7.1 Ablation

Table 4 compares Dipper with Dipper[⊕] and Dipper[∅]. The three ciphers were analysed under identical conditions: all 64 maximal cubes, MP-EL and BDP (which again agree), and the same solver settings (conflict budget 10^5 with a $5 \cdot 10^6$ retry; no instance was left unresolved). Within the model, the two additions shorten the longest certified property from nine rounds (XOR) or ten rounds (no addition) to six. The empirical word-cube experiment (Table 1) orders the three ciphers the same way: 5, 6 and 7 rounds. The data needed shifts in the same direction. For six rounds the smallest certified cubes found have size 2^{60} for Dipper, 2^{22} for Dipper[⊕] and 2^8 for Dipper[∅].

The variants keep the rotations, the S-box, the permutation and the full-state key addition, so they are not GIFT-64. A difference measured with an existence-based model describes certified properties. It describes the true algebraic behaviour only where the model is tight. We measured tightness only on Dipper for up to five rounds (Section 8).

7.2 Degree upper bounds

By Section 3, a bit of degree less than d is balanced over every cube of dimension d , so degree bounds give a global view of integral resistance. For every output bit j and round count r we computed the smallest D for which MP-EL with a free input mask of weight at least $D + 1$ and output e_j is unsatisfiable. This certifies the *upper bound* $\deg_x S_j^{(r)} \leq D$ for all round keys. Bounds are reported up to 63 (see Section 3); a bound of 63 carries no information about the actual degree. All 64×7 bounds for Dipper and 64×11 bounds for each variant were resolved without timeouts.

Figure 2 and Table 5 support three observations. All three concern upper bounds.

The addition acts immediately. After one round, the bound on bit i of $A \boxplus B$ grows almost linearly in i , from 3 to 26, the pattern predicted by Lemma 2 and Corollary 1. The retained word B stays at the S-box degree 2 or 3. The growth is slower than $3(i+1)$ because neighbouring bits share S-boxes.

Retained words lag by about one round. The mean bound on the added words reaches 60 at $r = 3$, and the retained words need one more round. After five rounds every added-word bit has bound 63, while six retained-word bits, $\{1, 19, 24, 51, 56, 58\}$, still have bound 62. A bound of 62 certifies the bit over every 63-dimensional cube. These six bits are exactly the bits certified

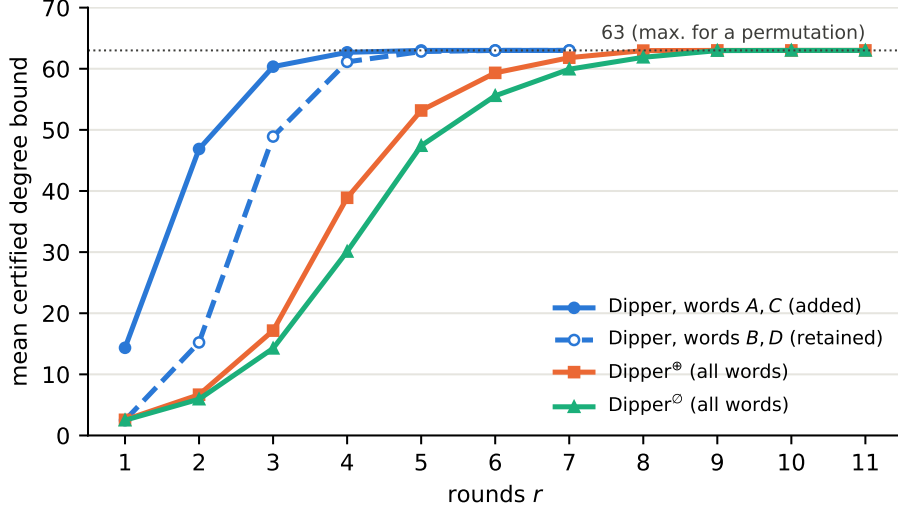


Figure 2: Mean certified *upper bound* on the algebraic degree of the output bits of $S^{(r)}$. For Dipper, bits are grouped by the word of the last ARX layer that produced them. Actual degrees may be lower.

Table 5: Certified degree upper bounds after one round, by bit position i in the output of the ARX layer.

i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$A \boxplus B$	3	5	7	8	9	11	13	15	16	18	20	21	22	24	25	26
$C \boxplus D$	2	4	6	6	8	10	12	12	14	16	18	18	20	22	24	24
B (retained)	2	2	3	3	2	2	3	3	2	2	3	3	2	2	3	3

for all 64 maximal cubes at five rounds, and the same six bits carry the six-round certificate of the 2^{60} cube (Table 3).

The bounds become uninformative three to four rounds earlier with the additions. From $r = 6$ on, every bound for Dipper is 63; for Dipper[⊕] and Dipper[∅] this happens only from $r = 9$ and $r = 10$ on (the last rounds with a bound below 63 are 5, 8 and 9, respectively). The ordering matches Table 4, but degree bounds do not determine the certificate boundary: at six rounds Dipper still has certificates for particular 63-dimensional cubes although every bound is 63, and Dipper[⊕] at eight rounds has 64 certified bits on some cube while only two bits have a bound below 63. Certificates are per cube, degree bounds are over all cubes (Remark 1). In the linear variants the bound can grow only through the S-boxes, at most by a factor of three per round. In Dipper each addition contributes the additional factor of Corollary 1. Whether the *actual* degree also reaches 63 earlier cannot be decided from upper bounds (Remark 1).

8 Precision of the models

MP-EL versus BDP. On all 64 maximal cubes, MP-EL and BDP produced identical certificate sets: for $r = 4, \dots, 9$ on Dipper, up to $r = 10$ on Dipper[⊕] and up to $r = 11$ on Dipper[∅]. MP-circuit agreed with both wherever it was run ($r \leq 7$). On 140 further (cube, round) instances of dimension 4–16 on Dipper, all three models produced identical certificate sets, at similar cost (8.4s, 8.8s and 6.2s in total). This shows that the two *existence* models agree on the tested instances. It does not show that cancellation-aware monomial prediction would have no advantage over BDP. Part of the agreement is structural. Every S-box layer of Dipper is

Table 6: Certified versus empirically zero bits over 28 cubes of dimension 4–16 (sums over all cubes).

Rounds	certified	empirically zero	gap	persistent gap
2	1283	1384	101	21
3	446	537	91	25
4	91	93	2	0
5	6	7	1	0

preceded by a full-state key addition. In MP-EL the composition “key addition, then S-box” allows exactly the transitions $u \rightarrow v$ with $S[w \rightarrow v] = 1$ for some $w \succeq u$, which is precisely the BDP S-box table. The two models can therefore differ only inside the ARX layer, and there we observed no difference.

Certificates versus experiment. Table 6 compares certificates with experiment on 28 cubes of dimension 4–16: word-aligned, nibble-aligned and random. A bit is *empirically zero* if its sum is zero in 200 trials, and it is a *gap* bit if it is empirically zero but not certified. Gap bits were re-tested with 3000 more trials (1000 for 16-dimensional cubes); *persistent* gaps stayed zero.

- At four and five rounds the certificates match experiment up to three gap bits, none of them persistent.
- At two and three rounds, about 7–17% of the empirically zero bits are not certified, and 46 bits stay zero in every re-test.

Such a bit is either an exact property lost to cancellation (Example 1) or a sum that is nonzero only with small probability. We tried two exact methods on these bits.

- *Trail parity per key monomial.* For bit 30 of a 12-dimensional random cube at two rounds, the number of trails exceeded our enumeration cap of 10^5 .
- *Low-weight evaluation.* The cube sum g is a multilinear polynomial in the key bits. If it has degree at most D , it vanishes identically iff it vanishes on every point of Hamming weight at most D . MP-EL gives the key support of g and a bound on D . For the 46 persistent gap bits the support has 33–88 key bits and the degree bound is 23–65. For 43 of them the number of low-weight points exceeds 10^{15} . The three smallest cases (support 33–38, degree bound 23–27) need about $8.5 \cdot 10^9$ – $2.7 \cdot 10^{11}$ points. Within a time limit of 800 s per case we evaluated 2.1–3.8 million points, covering all points of weight at most 5 or 6. No nonzero cube sum appeared, but the enumeration is far from complete, so none of these bits is decided.

We leave these cases open; a bitsliced implementation could complete the three smallest ones. In our sample they occur only at two and three rounds. The frontier itself (six and seven rounds, cubes of dimension 60–63) is out of experimental reach.

9 A partial-key filtering extension

This section describes how the six-round properties could be used for key filtering over one further round. We present it as a proposed procedure with an explicit cost model. It is not a validated eight-round attack.

Procedure. Let bit j of $S^{(6)}$ be certified over a cube I . For eight-round ciphertexts C , Lemma 4 gives

$$\bigoplus_{x \in \mathcal{C}_I(c)} T^{-1}(T^{-1}(C) \oplus RK_8)_j = 0.$$

Bit j of $T^{-1}(z)$ depends on four bits of z if its nibble lies in word B or D , where the inverse layer is only a rotation. It depends on up to 32 bits if the nibble lies in A or C , because of the borrow chain of the subtraction. For $j = 1$, which is certified over the 2^{60} cube, it depends on four bits of $Z = T^{-1}(C)$ and hence on four bits of RK_8 . One structure of 2^{60} chosen plaintexts is processed in a streaming fashion:

1. For each ciphertext, compute $Z = T^{-1}(C)$ (keyless) and toggle one of 16 parity bits indexed by the four relevant bits of Z .
2. Let $h : \mathbb{F}_2^4 \rightarrow \mathbb{F}_2$ be the partial inverse that maps the four relevant bits of z to bit j of $T^{-1}(z)$ (inverse rotation of word D , then the inverse S-box of the corresponding nibble). For each of the 16 guesses g , evaluate $\bigoplus_{v: \text{parity}[v]=1} h(v \oplus g)$ over the 16 table entries, and discard g if the result is 1.

One structure costs 2^{60} evaluations of T^{-1} plus 2^8 table operations. It needs 16 bits of memory besides the ciphertext stream. Every structure uses a fresh value of the four constant bits.

Assumptions and small-scale evidence. The right guess always passes. The number of wrong guesses left after N structures depends on how the wrong-guess sums behave. Under the usual heuristic that they are independent and uniform, N structures leave $15 \cdot 2^{-N}$ wrong guesses on average. We ran the same procedure at small scale, on six rounds: the 4-round certificate of the cube $\{60, \dots, 63\}$ on bit 1, followed by the free round and one key-guessed round, with 200 random keys. The right guess always survived. A wrong guess survived one structure with probability about 0.565, so the uniform heuristic is optimistic, and eight structures left 1.13 candidates on average, with a unique survivor in 88.5% of the runs. We have not established that this rate carries over to eight rounds or that the structures act as independent filters there. Extrapolating anyway, about eight structures, i.e. 2^{63} chosen plaintexts, would be needed to identify four bits of RK_8 . This recovers round-key bits, not the master key, and the data requirement is half the codebook. We therefore regard the procedure as evidence of a large margin, not as an attack on Dipper.

10 Specification of the analysed cipher

The results of this paper depend only on the round function, because every certificate holds for arbitrary independent round keys. The round function is fixed by Section 3 and confirmed by the published Dipper-64/128 test vectors, which exercise all 28 rounds. Appendix A lists the vectors together with intermediate round keys and states.

Two discrepancies with the key-schedule text of [21] were found.

1. The Dipper-64/128 vectors are reproduced when the round key is $RK_r = K^{(r)}[63:0]$, while Equation (7) of [21] states $K^{(r)}[127:64]$.
2. The two Dipper-64/96 vectors are not reproduced by the key schedule as written. They are also not reproduced by any variant in a brute-force search that covered:
 - all 720 word permutations of the key update;
 - all placements and directions of the two word rotations;
 - every 3- or 4-subset of S-box words;

- all six round-constant positions;
- three extraction windows.

[To be completed by the authors before submission: the Dipper-64/96 key schedule as implemented in the reference code, its test vectors, intermediate round keys, and the corresponding correction to [21].]

11 Discussion

Security implications. The integral properties certified here reach six of Dipper’s 28 rounds on the state, and seven with the keyless final-round inversion. The filtering extension of Section 9 would need half the codebook to reach eight rounds. Integral attacks therefore leave a large margin. This is consistent with the differential, linear and impossible-differential bounds of [21]. None of these results is a proof of security.

Design implication. The ablation and the degree bounds point the same way. With the two half-state additions, the certifiable integral properties end three to four rounds earlier than for an otherwise identical round without carries. Within the state, the retained operands are the part whose degree bound grows most slowly, and every certified property at the boundary lives there. A designer who wants integral saturation to happen sooner could therefore target the retained words, for example by alternating which words are added in consecutive rounds. We have not evaluated this change.

Towards exact bounds. The main open question is the gap between “no certificate” and “no distinguisher”. Closing it requires proving that monomials are present, for each relevant cube separately (Remark 1). Hebborn et al. [15, 16] prove presence by finding a key monomial with an odd number of trails; Zeng and Tian [40] adapt the method to ciphers without a whitening key. We tried a direct version on two seven-round, 63-dimensional instances: take the key monomial of one trail, then count all trails with that monomial. With 20 random key monomials per instance we found no odd count within the enumeration limit of $2 \cdot 10^4$ trails. A tailored choice of key monomials, as in [16], is the most promising route.

Limitations.

- The seven-round statement holds within an existence-based model. It does not exclude a seven-round property that depends on trail cancellation. It also says nothing about input sets that are not bit-aligned cubes, or about output functions other than single bits.
- Independent round keys make the certificates stronger, since they hold for all keys. For the same reason they may miss properties that depend on the key schedule.
- The data complexities are those of the smallest certified cubes found by a greedy search.
- The gap bits of Section 8 are unresolved.
- Degree results are upper bounds.
- The filtering extension relies on an extrapolated survival rate.

12 Conclusion

The experimental five-round integral bound of Dipper is now a key-independent result, backed by checked DRAT proofs. Certified properties extend to six rounds, with 2^{60} data for the smallest cube found, and to seven rounds with the keyless final-round inversion. Within the existence-based model, no bit-aligned cube reaches seven rounds, and every trail behind this statement has been validated independently. The two half-state modular additions account for the difference between six certified rounds and the nine or ten rounds of otherwise identical rounds without carries. Only retained-operand bits are certified at the boundary. This is consistent with the exact degree of the addition output bits and with the degree upper bounds for every output bit and round, which grow fastest on the added words. On every Dipper instance we examined, the existence model with exact local transitions certifies the same bits as the conventional division property, partly because a full-state key addition precedes each S-box layer. Separating “no certificate” from “no distinguisher” at seven rounds remains open.

Data and code availability

The reference implementation, SAT models, scripts, raw JSON results, the proof-checking pipeline and the sources of this paper are available at <https://github.com/AlexGuseinov/dipper-integral>. The CNF files and DRAT proofs (about 60 MB) are provided as a release archive of the repository. `make test` runs all validation checks, `make reproduce` regenerates every table and figure, and `make certificates` regenerates and checks all proofs and trails.

A Test vectors and intermediate values

Dipper-64/128 under the convention $RK_r = K^{(r)}[63:0]$ (Section 10); all values in hexadecimal, most significant bit first.

	Vector 1	Vector 2
K	000102030405060708090A0B0C0D0E0F	FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
P	0123456789ABCDEF	FFFFFFFFFFFFFFFF
RK_1	08090A0B0C0D0E0F	FFFFFFFFFFFFFFFF
$S^{(1)}$	87D2D72CD66C68F2	3118311831183108
RK_2	140506076303E0F0	EEEEEEEEEEEEEEEE
$S^{(2)}$	41C18243B4605E18	01A6DC1CB8EEB9D1
RK_3	1001020208C00F0E	EEEEEEFEEBFFFF
$S^{(3)}$	22C358EDC3BD1695	F49F4481BC92B880
RK_{28}	78090A148A9BF0E0	FFFFFEE02664FFFF
$C = S^{(28)}$	4B46284387969060	6ABB4518063706B0

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